Lecture 2

Exponential maps and local group structure

Objectives. *A tangent vector at the identity generates a one-parameter subgroup. The exponential map collects these subgroups into one map from the Lie algebra to the group. We ask how much of the multiplication is determined by the bracket. Near the identity, the bracket determines the whole multiplication. Far from the identity, it does not.*

By the end of this lecture you will be able to:

- Compute exponentials of rotation and boost generators and of elements of $\mathfrak{sl}_2(\mathbb{R})$.
- Identify continuous one-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$ with exponentials.
- Differentiate a homomorphism, and use $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ and $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\,X}$.
- Compute the second-order BCH term and recognise non-exponential elements.

2.1 The matrix exponential

For a complex $n \times n$ matrix X, let $\|X\|$ be the operator norm for the standard Hermitian norm on $\mathbb{C}^n$. The operator norm satisfies $\|XY\| \le \|X\|\,\|Y\|$.

Definition 2.1 (Matrix exponential)**.** For $X \in M_{n\times n}(\mathbb{C})$, the exponential of X is

$$\exp X = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

Since $\|X^k\| \le \|X\|^k$, the series is dominated by the scalar series of $e^{\|X\|}$. Hence the series converges absolutely, and uniformly on bounded sets. Each entry of $\exp X$ is an everywhere convergent power series in the entries of X. So exp is smooth, and its series may be differentiated term by term.

Proposition 2.2 (Properties of the matrix exponential)**.** *Let $X, Y \in M_{n\times n}(\mathbb{C})$ and $g \in \mathrm{GL}_n(\mathbb{C})$.*

(i) *If $XY = YX$, then $\exp(X+Y) = \exp X \exp Y$.*

(ii) *The matrix $\exp X$ is invertible, with inverse $\exp(-X)$.*

(iii) *The curve $t \mapsto \exp(tX)$ has derivative $X\exp(tX) = \exp(tX)X$.*

(iv) $g\exp(X)g^{-1} = \exp(gXg^{-1})$.

(v) $\det\exp X = e^{\mathrm{tr}\,X}$.

Proof. (i) Absolute convergence lets us multiply the series and collect terms of total degree m. The binomial theorem holds because $XY = YX$:

$$\exp X \exp Y = \sum_{m=0}^{\infty} \sum_{k+l=m} \frac{X^k Y^l}{k!\,l!} = \sum_{m=0}^{\infty} \frac{(X+Y)^m}{m!}.$$

(ii) Apply the product rule (i) to the commuting pair X and $-X$. (iii) Differentiate $\sum_k t^k X^k/k!$ term by term and factor out X on either side. (iv) We have $(gXg^{-1})^k = gX^k g^{-1}$. Conjugation by g is linear and continuous, so it commutes with the sum of the series. (v) Schur triangularisation gives $X = PTP^{-1}$ with T upper triangular. Let t_{jj} be the diagonal entries of T. Then $\exp T$ is upper triangular with diagonal entries $e^{t_{jj}}$. By the conjugation formula (iv),

$$\det \exp X = \det \exp T = \prod_j e^{t_{jj}} = e^{\operatorname{tr} T} = e^{\operatorname{tr} X}.$$ ∎

The hypothesis $XY = YX$ in the product rule, Proposition 2.2(i), cannot be dropped. Exercise 2.3 gives nilpotent matrices E and F with $\exp E \exp F \neq \exp(E+F)$. The Baker–Campbell–Hausdorff formula in § 2.5 shows that the bracket governs the correction.

2.2 One-parameter subgroups

Definition 2.3 (One-parameter subgroup)**.** A one-parameter subgroup of a Lie group G is a continuous homomorphism $\varphi\colon \mathbb{R} \to G$ from the additive group of real numbers.

For every matrix X, the curve $t \mapsto \exp(tX)$ is a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$, by the product rule, Proposition 2.2(i). Conversely, continuity alone forces this form.

Theorem 2.4 (One-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$)**.** *Let $\varphi\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{C})$ be a one-parameter subgroup. Then φ is differentiable, and $\varphi(t) = \exp(tX)$ for the unique matrix $X = \varphi'(0)$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then X is a real matrix.*

Proof. We first show that φ is differentiable. Averaging a continuous function over a moving interval gives a differentiable function. The homomorphism law writes φ as its own moving average times a fixed matrix.

Step 1: differentiability. By continuity of φ, choose $\varepsilon > 0$ with $\|\varphi(s) - I\| \le \frac{1}{2}$ for $0 \le s \le \varepsilon$. Integrate entrywise and set

$$M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(s)\, ds.$$

Then $\|M - I\| \le \frac{1}{2}$, so the Neumann series $\sum_{k\ge 0} (I-M)^k$ inverts M. The homomorphism property gives

$$\varphi(t) M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(t+s)\, ds = \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \varphi(u)\, du.$$

The right side is continuously differentiable in t, by the fundamental theorem of calculus. Multiplication by the constant matrix M^{-1} preserves this property, so φ is continuously differentiable too.

Step 2: the equation. Put $X = \varphi'(0)$. Differentiate $\varphi(t+s) = \varphi(t)\varphi(s)$ in s at $s = 0$. This gives $\varphi'(t) = \varphi(t) X$.

Step 3: uniqueness. Let $\psi(t) = \varphi(t)\exp(-tX)$. By the Leibniz rule and the derivative formula, Proposition 2.2(iii),

$$\psi'(t) = \varphi(t)X\exp(-tX) - \varphi(t)X\exp(-tX) = 0.$$

Hence $\psi(t) = \psi(0) = I$, and $\varphi(t) = \exp(tX)$. Any Y with $\varphi(t) = \exp(tY)$ satisfies $Y = \varphi'(0) = X$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then its derivative $X = \varphi'(0)$ is a real matrix, as a limit of real difference quotients. ∎

Example 2.5 (Rotations)**.** The algebra $\mathfrak{so}(2)$ is spanned by

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Separating even and odd powers in the series gives

$$\exp(\theta J) = \cos\theta\, I + \sin\theta\, J = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The first column of an element of $\mathrm{SO}(2)$ is a unit vector, so it is $(\cos\theta, \sin\theta)$ for some θ. Orthogonality and determinant one then fix the second column. So $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ is onto. It is not injective, since $\exp(2\pi J) = I$.

Example 2.6 (Boosts)**.** In special relativity on a line, with the speed of light set to 1, an event has a time coordinate x_0 and a position x_1. Observers in uniform relative motion use coordinates related by linear maps preserving $x_0^2 - x_1^2$. With $\eta = \mathrm{diag}(1, -1)$, these maps form the group

$$\mathrm{O}(1,1) = \{g \in M_{2\times 2}(\mathbb{R}) : g^{\mathsf{T}}\eta g = \eta\}.$$

Let $g(t)$ be a smooth curve in $\mathrm{O}(1,1)$ with $g(0) = I$ and $g'(0) = X$. Differentiating $g(t)^{\mathsf{T}}\eta g(t) = \eta$ at $t = 0$ gives the necessary condition $X^{\mathsf{T}}\eta + \eta X = 0$. In terms of the entries of X,

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad X^{\mathsf{T}}\eta + \eta X = \begin{pmatrix} 2a & b-c \\ b-c & -2d \end{pmatrix}.$$

The condition therefore forces $a = d = 0$ and $b = c$. So X is a multiple of

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

The even and odd parts of the series now give hyperbolic functions:

$$B(u) = \exp(uK) = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix}.$$

Since $\cosh^2 u - \sinh^2 u = 1$, we get $B(u)^{\mathsf{T}}\eta B(u) = \eta$. So K is the velocity of a curve in the group, and the tangent algebra is exactly $\mathbb{R}K$.

The parameter u is the rapidity. The boost $B(u)$ sends the events $(\tau, 0)$ of an observer at rest to $(\tau\cosh u, \tau\sinh u)$, a motion with velocity $\tanh u$. Rapidities add, since $B(u)B(u') = B(u+u')$. Velocities do not add. Unlike the parametrisation of rotations, the map $u \mapsto B(u)$ is injective. Every boost has trace $2\cosh u \geq 2$.

2.3 The exponential map of a Lie group

So far our groups are matrix groups. Not every Lie group is one, and the reading of Lecture 4 cites an example. On a general Lie group, the exponential power series is not defined. Instead we use the left-invariant vector fields $X^L(g) = (dL_g)_e X$ of Lecture 1, with $X \in \mathfrak{g} = T_eG$.

Definition 2.7 (Integral curve). An integral curve of a vector field V on a manifold M is a smooth curve $\gamma : I \to M$, on an open interval I, with $\gamma'(t) = V(\gamma(t))$ for all $t \in I$.

In a chart, this condition is a first-order system of ODEs with smooth coefficients. The Picard–Lindelöf theorem of the ODE background gives local existence and uniqueness. Two integral curves with the same initial point agree on their common interval. The agreement set is open by local uniqueness, closed because limits in a Hausdorff manifold are unique, and nonempty. An interval is connected, so it is the whole interval.

Left translation maps integral curves of X^L to integral curves of X^L. For such a curve γ and $g \in G$, the chain rule and $L_g \circ L_h = L_{gh}$ give

$$(L_g \circ \gamma)'(t) = (dL_g)_{\gamma(t)}(dL_{\gamma(t)})_e X = (dL_{g\gamma(t)})_e X = X^L\big(g\gamma(t)\big).$$

Theorem 2.8 (Exponential map of a Lie group; cited). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $X \in \mathfrak{g}$. The integral curve γ_X of X^L with $\gamma_X(0) = e$ is defined on all of $\mathbb{R}$. The map*

$$\exp : \mathfrak{g} \longrightarrow G, \qquad \exp X = \gamma_X(1),$$

is smooth, and $\gamma_X(t) = \exp(tX)$ for all real t.

This is proved in Kirillov Jr., Chapter 3, in the section on the exponential map, and in Warner, Chapter 3. We sketch the algebraic part. The omitted, cited step is that γ_X exists for all t and that $\gamma_X(1)$ depends smoothly on X.

For fixed s, the curves $t \mapsto \gamma_X(s+t)$ and $t \mapsto \gamma_X(s)\gamma_X(t)$ are integral curves of X^L starting at $\gamma_X(s)$. By uniqueness,

$$\gamma_X(s+t) = \gamma_X(s)\gamma_X(t).$$

For a real number c, the curve $t \mapsto \gamma_X(ct)$ is the integral curve of $(cX)^L$ through e. So $\gamma_{cX}(t) = \gamma_X(ct)$. At $t = 1$ this gives $\exp(cX) = \gamma_X(c)$.

Let $G \subseteq \mathrm{GL}_n(\mathbb{C})$ be an embedded matrix Lie group, as in Lecture 1. Left translation is matrix multiplication, so γ_X solves $\gamma' = \gamma X$ with $\gamma(0) = I$ in the matrix space. Step 3 of the proof of the one-parameter subgroup theorem, Theorem 2.4, shows that $\exp(tX)$ is the only solution. Hence the two exponentials agree, and $\exp(tX) \in G$ for every $X \in \mathfrak{g}$.

Lemma 2.9 (Smooth one-parameter subgroups). *Let $\beta : \mathbb{R} \to G$ be a smooth homomorphism with $\beta'(0) = X$. Then $\beta(t) = \exp(tX)$ for all t.*

Proof. Differentiate $\beta(t+s) = L_{\beta(t)}\beta(s)$ in s at $s = 0$. This gives $\beta'(t) = X^L(\beta(t))$. A homomorphism sends the identity to the identity, so $\beta(0) = e$. Hence β and γ_X are both integral curves of X^L through e. By uniqueness of integral curves, they agree. ∎

Proposition 2.10 (Local invertibility)**.** *The differential of* exp *at* 0 *is the identity of* $\mathfrak{g}$*. Consequently there are open sets* $U \ni 0$ *in* $\mathfrak{g}$ *and* $V \ni e$ *in* G *such that* exp *restricts to a diffeomorphism* $U \to V$*. We may take* $U = -U$*.*

Proof. Identify $T_0\mathfrak{g}$ with $\mathfrak{g}$. The line $t \mapsto tX$ has velocity X. Its image under exp is γ_X, whose velocity at 0 is $X^L(e) = X$. Hence $d\exp_0(X) = X$.

Read exp in a chart around e. The inverse function theorem of multivariable calculus, which we cite, says the following. A smooth map between open subsets of $\mathbb{R}^d$ with invertible derivative at a point restricts to a diffeomorphism between open neighbourhoods of that point and its image. This gives U and V. Replace U by $U \cap (-U)$, and replace V by the image of $U \cap (-U)$ under exp. Both sets remain open. ∎

Write $\log\colon V \to U$ for the inverse. Local invertibility, Proposition 2.10, is a local statement. Indeed, the rotation example, Example 2.5, shows that exp need not be injective on all of $\mathfrak{g}$.

Proposition 2.11 (Generation by exponentials)**.** *If* G *is connected, every element of* G *is a finite product* $\exp X_1 \exp X_2 \cdots \exp X_k$*.*

Proof. Let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. Since $\exp(-X) = (\exp X)^{-1}$ and $U = -U$, we have $V^{-1} = V$. Let H be the set of finite products of elements of V. It is a subgroup. It is open, since $H = \bigcup_{h \in H} hV$. Its complement is a union of cosets gH, so it is open too. As $e \in H$ and G is connected, $H = G$. ∎

The proof uses only that V is open, contains e and satisfies $V^{-1} = V$.

A group that is not connected still has a connected part around e. The next definition describes it by paths.

Definition 2.12 (Identity component)**.** The *identity component* G_0 of a Lie group G is the set of elements of G joined to e by a continuous path in G.

For example, let $G = \mathrm{O}(2)$. The determinant is continuous with values ± 1, so it is constant along a path. Hence $G_0 \subseteq \mathrm{SO}(2)$. Conversely, every element of $\mathrm{SO}(2)$ is $\exp(\theta J)$ by the rotation example, Example 2.5, and $t \mapsto \exp(t\theta J)$ joins it to I. So $G_0 = \mathrm{SO}(2)$.

Proposition 2.13 (Properties of the identity component)**.** *Let* G *be a Lie group with Lie algebra* $\mathfrak{g}$*.*

(a) G_0 *is a normal subgroup of* G*, and it is both open and closed.*

(b) G_0 *has Lie algebra* $\mathfrak{g}$*, and it contains* $\exp(tX)$ *for all* $X \in \mathfrak{g}$ *and* $t \in \mathbb{R}$*.*

(c) G_0 *is connected, and it contains every connected subset of* G *through* e*. Hence* G *is connected exactly when* $G = G_0$*.*

Proof. (a) Let γ and δ be continuous paths from e to g and from e to h. Then $t \mapsto \gamma(t)\delta(t)$ joins e to gh, and $t \mapsto \gamma(t)^{-1}$ joins e to g^{-1}. For $x \in G$, the path $t \mapsto x\gamma(t)x^{-1}$ joins e to xgx^{-1}. So G_0 is a normal subgroup.

Next, let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. For $g \in G_0$ and $X \in U$, the path $t \mapsto g\exp(tX)$, $0 \le t \le 1$, joins g to $g\exp X$. Hence the open set

gV lies in G_0, so G_0 is open. Its complement is a union of cosets gG_0. These cosets are open, so G_0 is closed.

(b) An open subset containing e has the same tangent space at e as G. The curve $s \mapsto \exp(sX)$ joins e to $\exp(tX)$.

(c) Suppose $G_0 = O_1 \cup O_2$ with O_1 and O_2 disjoint, open in G_0 and nonempty, and with $e \in O_1$. Choose $g \in O_2$ and a path γ from e to g in G. Every point of this path is joined to e along γ, so γ lies in G_0. Then $\gamma^{-1}(O_1)$ and $\gamma^{-1}(O_2)$ would disconnect $[0,1]$. So G_0 is connected.

Let C be a connected subset of G containing e. By (a), $C \cap G_0$ is open and closed in C, and it is nonempty. So $C \subseteq G_0$. Thus G_0 is the connected component of e. Finally, if G is connected, then taking $C = G$ gives $G = G_0$. The converse is the first statement of this part. ∎

2.4 Conjugation and homomorphisms

Conjugation $C_g(h) = ghg^{-1}$ is a smooth automorphism of G fixing e. The adjoint action of g is the linear map $\mathrm{Ad}_g = (dC_g)_e \colon \mathfrak{g} \to \mathfrak{g}$, also written $\mathrm{Ad}(g)$. For a matrix group, differentiating $g\exp(tX)g^{-1}$ at $t = 0$ gives $\mathrm{Ad}_g X = gXg^{-1}$. Hence $gXg^{-1} \in \mathfrak{g}$.

Theorem 2.14 (Differentiating homomorphisms). *Let $\Phi \colon G \to H$ be a Lie group homomorphism and $\phi = d\Phi_e$.*

(i) $\Phi(\exp X) = \exp(\phi(X))$ *for all* $X \in \mathfrak{g}$.

(ii) $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ *for all* $g \in G$ *and* $X \in \mathfrak{g}$.

(iii) If G is connected, then Φ is determined by ϕ.

(iv) If G and H are matrix Lie groups, then $\phi([X,Y]) = [\phi(X), \phi(Y)]$.

Proof. (i) The curve $\beta(t) = \Phi(\exp tX)$ is a smooth homomorphism $\mathbb{R} \to H$ with velocity $\phi(X)$ at 0. Apply the lemma on smooth one-parameter subgroups, Lemma 2.9, in H and set $t = 1$.

(ii) Apply the intertwining formula (i) to $\Phi = C_g$, whose differential is Ad_g.

(iii) By generation by exponentials, Proposition 2.11, each element of G is a product $\exp X_1 \cdots \exp X_k$. By the intertwining formula (i), the image of this product under Φ is $\exp \phi(X_1) \cdots \exp \phi(X_k)$, which depends only on ϕ.

(iv) For $g \in G$ and $Y \in \mathfrak{g}$, the matrix $gYg^{-1} \in \mathfrak{g}$ is the velocity of $g\exp(tY)g^{-1}$ at $t = 0$. Differentiate $\Phi(g\exp(tY)g^{-1}) = \Phi(g)\Phi(\exp tY)\Phi(g)^{-1}$ at $t = 0$:

$$\phi(gYg^{-1}) = \Phi(g)\,\phi(Y)\,\Phi(g)^{-1}.$$

Put $g = \exp(sX)$ and use the intertwining formula (i) on the right:

$$\phi\big(e^{sX}Ye^{-sX}\big) = e^{s\phi(X)}\phi(Y)e^{-s\phi(X)}.$$

Differentiate at $s = 0$, using the derivative formula, Proposition 2.2(iii). Since ϕ is linear, it commutes with differentiation. The result is $\phi(XY - YX) = \phi(X)\phi(Y) - \phi(Y)\phi(X)$. This is the bracket identity $\phi([X,Y]) = [\phi(X), \phi(Y)]$. ∎

The bracket identity, Theorem 2.14(iv), holds for every Lie group homomorphism. The general proof uses the vector-field bracket of Lecture 1. We cite Kirillov Jr., Chapter 3.

The most important target is $H = \mathrm{GL}(V)$, for a finite-dimensional complex vector space V. A representation $\Pi\colon G \to \mathrm{GL}(V)$ has a differential $d\Pi$ at e, and the intertwining formula (i) reads $\Pi(\exp X) = \exp(d\Pi(X))$.

We now apply the theorem to the adjoint action itself. Here $\mathrm{GL}(\mathfrak{g})$ is the group of invertible linear maps of $\mathfrak{g}$, and $\mathrm{End}(\mathfrak{g})$ is its tangent space at the identity.

Proposition 2.15 (The adjoint representation)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$. The map $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism. Write $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ for its differential at e. Then, for all $X, Y \in \mathfrak{g}$,*

$$\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\,X}, \qquad \mathrm{ad}\,X(Y) = [X, Y].$$

Proof. Since $C_{gh} = C_g \circ C_h$, the chain rule gives $\mathrm{Ad}_{gh} = \mathrm{Ad}_g\,\mathrm{Ad}_h$. The map $(g, h) \mapsto ghg^{-1}$ is smooth, so its derivative in h at $h = e$ depends smoothly on g. Hence Ad is a smooth homomorphism.

The intertwining formula, Theorem 2.14(i), with $\Phi = \mathrm{Ad}$ and $H = \mathrm{GL}(\mathfrak{g})$, gives $\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}\,X)$. This is the exponential of the operator $\mathrm{ad}\,X$, which we write as $e^{\mathrm{ad}\,X}$.

By the definition of a differential, $\mathrm{ad}\,X(Y)$ is the derivative of $\mathrm{Ad}(\exp sX)Y$ at $s = 0$. For a matrix group, $\mathrm{Ad}(\exp sX)Y = e^{sX}Ye^{-sX}$, and the derivative formula, Proposition 2.2(iii), gives $\mathrm{ad}\,X(Y) = XY - YX = [X, Y]$. For a general Lie group we cite $\mathrm{ad}\,X(Y) = [X, Y]$ from Warner, Chapter 3, Section 3.46 and Proposition 3.47. ∎

For example, in $\mathrm{GL}_2(\mathbb{R})$ take

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad [h, E] = 2E.$$

So $\mathrm{ad}\,h$ multiplies E by 2, and $e^{t\,\mathrm{ad}\,h}E = e^{2t}E$. Direct multiplication with $\exp(th) = \mathrm{diag}(e^t, e^{-t})$ gives the same result: $\exp(th)E\exp(-th) = e^{2t}E$. Exercise 2.5 computes the operator $\mathrm{ad}\,A$ on $\mathfrak{su}(2)$.

The adjoint representation gives the product rule for commuting elements of any Lie group.

Proposition 2.16 (Commuting exponentials)**.** *Let G be a Lie group, and let $X, Y \in \mathfrak{g}$ with $[X, Y] = 0$. Then $\exp(tX)$ and $\exp(sY)$ commute for all $s, t \in \mathbb{R}$, and*

$$\exp(X + Y) = \exp X \exp Y.$$

Proof. Since $\mathrm{ad}\,X(Y) = [X, Y] = 0$, we get $\mathrm{Ad}(\exp tX)Y = e^{t\,\mathrm{ad}\,X}Y = Y$. Apply Theorem 2.14(ii) with $g = \exp(tX)$ to the element sY:

$$\exp(tX)\exp(sY)\exp(-tX) = \exp\bigl(s\,\mathrm{Ad}(\exp tX)Y\bigr) = \exp(sY).$$

So $\exp(tX)$ and $\exp(sY)$ commute. Hence $\beta(t) = \exp(tX)\exp(tY)$ is a smooth homomorphism $\mathbb{R} \to G$.

Its velocity at 0 is $X+Y$. Indeed, let $m(g,h)=gh$. Since $m(g,e)=g$ and $m(e,h)=h$, the differential of m at (e,e) sends $(X,0)$ to X and $(0,Y)$ to Y. By linearity it sends (X,Y) to $X+Y$. The lemma on smooth one-parameter subgroups, Lemma 2.9, gives $\beta(t)=\exp\bigl(t(X+Y)\bigr)$. Now put $t=1$. ∎

The same differential shows that the velocity at 0 of a product of curves through e is the sum of their velocities.

2.5 The Baker–Campbell–Hausdorff formula

By local invertibility, Proposition 2.10, a product of exponentials of small elements has a logarithm in $\mathfrak{g}$. We compute its leading terms in $\mathrm{GL}_n(\mathbb{C})$. Write $O(t^k)$ for a quantity bounded in norm by a constant times $|t|^k$ for small $|t|$.

Proposition 2.17 (Second-order term)**.** *Let $X,Y\in M_{n\times n}(\mathbb{C})$. For small $|t|$, put $Z(t)=\log\bigl(\exp(tX)\exp(tY)\bigr)$. Then*

$$Z(t)=t(X+Y)+\frac{t^2}{2}[X,Y]+O(t^3).$$

Proof. The map Z is smooth with $Z(0)=0$. Taylor's theorem gives matrices A,B with $Z(t)=tA+t^2B+O(t^3)$. Hence $\|Z(t)\|=O(t)$, so the tail $\sum_{k\ge3}Z^k/k!$ has norm at most $\|Z\|^3e^{\|Z\|}=O(t^3)$. Therefore

$$\exp Z(t)=I+Z+\tfrac12Z^2+O(t^3)=I+tA+t^2\bigl(B+\tfrac12A^2\bigr)+O(t^3).$$

Multiplying the two exponential series to second order gives

$$\exp(tX)\exp(tY)=I+t(X+Y)+t^2\bigl(\tfrac12X^2+XY+\tfrac12Y^2\bigr)+O(t^3).$$

The two left sides are equal. If $tP+t^2Q=O(t^3)$, dividing by t and letting $t\to0$ gives $P=0$, and then $Q=0$. Comparing coefficients,

$$A=X+Y,\qquad B=\tfrac12X^2+XY+\tfrac12Y^2-\tfrac12(X+Y)^2=\tfrac12(XY-YX).$$ ∎

The second-order term also shows what the bracket means. Multiplying out the series gives

$$\exp(tX)\exp(tY)\exp(-tX)\exp(-tY)=I+t^2[X,Y]+O(t^3),$$

as Exercise 2.6 shows. So the bracket measures, to leading order, how far $\exp(tX)$ and $\exp(tY)$ fail to commute. On a general Lie group, the flow of X^L is right multiplication by $\exp(tX)$. So this product is the end point of the path from e that follows X, Y, $-X$ and $-Y$ in turn. Figure 2.1 draws this path.

Exercise 2.7 computes the third-order term. The general statement is the following.

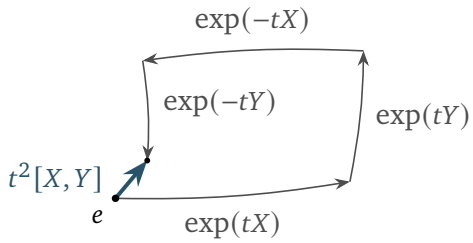


Figure 2.1: The path from e with steps $\exp(tX)$, $\exp(tY)$, $\exp(-tX)$ and $\exp(-tY)$, each multiplying on the right. The coloured arrow goes from e to the end point. In the coordinates given by log, this displacement is $t^2[X,Y] + O(t^3)$. The drawing is schematic.

Theorem 2.18 (Convergent BCH formula; cited)**.** *Let G be a Lie group with finite-dimensional real Lie algebra $\mathfrak{g}$. There is a neighbourhood W of 0 in $\mathfrak{g}$ with the following property. For $X, Y \in W$, the series grouped by total degree,*

$$H(X,Y) = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) + \cdots,$$

converges in $\mathfrak{g}$, depends continuously on (X,Y), and satisfies $\exp X \exp Y = \exp H(X,Y)$. The degree-$m$ term is a combination of iterated brackets of m entries from X and Y, with universal rational coefficients independent of G.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves this convergent series for connected groups. The identity component reduces a general G to that case, by Proposition 2.13. The section on sources at the end of this lecture records what Hall, Knapp and Kirillov Jr. each cover.

The theorem answers the main question near the identity. As before, $\log\colon V \to U$ is the local inverse of exp on G.

Proposition 2.19 (Local determination)**.** *Let G and G' be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{g}'$. Write $\exp'$ and e' for the exponential and the identity of G'. Let $\psi\colon \mathfrak{g} \to \mathfrak{g}'$ be a homomorphism of real Lie algebras, and put $f = \exp' \circ \psi \circ \log$ on V. Then f is smooth and $df_e = \psi$. There is an open neighbourhood $V_1 \subseteq V$ of e such that, for all $a, b \in V_1$,*

$$ab \in V \quad \text{and} \quad f(ab) = f(a)f(b).$$

If ψ is an isomorphism, then f restricts to a diffeomorphism from an open neighbourhood of e onto an open neighbourhood of e'.

Proof. Since ψ is linear and preserves brackets, it maps each term of $H(X,Y)$ to the corresponding term of $H(\psi X, \psi Y)$. Since it is continuous, it maps the convergent sum to the sum.

Write W' for the BCH neighbourhood of G'. Choose V_1 so close to e that, for all $a, b \in V_1$, the elements $X = \log a$ and $Y = \log b$ satisfy three conditions:

(a) $X, Y \in W$;
(b) $\psi X, \psi Y \in W'$;
(c) $H(X,Y) \in U$, the set on which exp is inverted.

Continuity of log, ψ and H, with $H(0,0)=0$, makes this possible. Since $X, Y \in W$ by condition (a), $ab = \exp H(X,Y)$. Since $H(X,Y) \in U$ by condition (c), $ab \in V$ and $\log(ab) = H(X,Y)$. Therefore

$$f(ab) = \exp'\big(\psi H(X,Y)\big) = \exp'\big(H(\psi X, \psi Y)\big) = \exp'(\psi X)\exp'(\psi Y) = f(a)f(b).$$

The third equality uses condition (b), $\psi X, \psi Y \in W'$.

The map f is smooth, as a composite of smooth maps. We have $f(e) = \exp'(\psi(0)) = \exp'(0) = e'$. By local invertibility, Proposition 2.10, $d\exp_0$ and $d\exp'_0$ are identity maps, so $d\log_e$ is the identity too. The chain rule gives

$$df_e = d\exp'_0 \circ \psi \circ d\log_e = \psi.$$

If ψ is an isomorphism, it is invertible. The inverse function theorem then gives an open neighbourhood $V_0 \subseteq V$ of e that f maps diffeomorphically onto an open neighbourhood of e'. ∎

So isomorphic Lie algebras give the same multiplication near the identity. The multiplicative part of the proposition does not need ψ to be injective. Lecture 4 uses the same argument for a homomorphism that need not be injective.

The bracket does not determine the global group. For example, $\mathrm{O}(n)$ and $\mathrm{SO}(n)$ have the same Lie algebra. The next section shows a subtler effect.

2.6 Exponentials in $\mathrm{SL}(2,\mathbb{R})$

We now show that exp need not be onto, even on a connected group.

For $g \in \mathrm{SL}(2,\mathbb{R})$, expanding a 2×2 determinant gives

$$\det(g + tM) = 1 + t\,\mathrm{tr}(g^{-1}M) + t^2 \det(g^{-1}M).$$

Hence $d\det_g(M) = \mathrm{tr}(g^{-1}M)$. This differential is onto $\mathbb{R}$, since $d\det_g(g) = \mathrm{tr}\, I = 2$. By the regular level-set theorem of Lecture 1, $\mathrm{SL}(2,\mathbb{R})$ is an embedded matrix Lie group. Its algebra $\mathfrak{sl}_2(\mathbb{R})$ is the kernel of $d\det_I$, so it consists of the real traceless 2×2 matrices.

Let $X \in \mathfrak{sl}_2(\mathbb{R})$. Cayley–Hamilton gives $X^2 = \delta I$ with $\delta = -\det X$. Separating even and odd powers, as for rotations and boosts,

$$\exp X = c(\delta) I + s(\delta) X, \qquad c(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k)!}, \quad s(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k+1)!}.$$

Comparison with the hyperbolic and circular series gives

$$c(\delta) = \begin{cases} \cosh\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \cos\sqrt{-\delta}, & \delta < 0, \end{cases} \qquad s(\delta) = \begin{cases} \sinh\sqrt{\delta}/\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \sin\sqrt{-\delta}/\sqrt{-\delta}, & \delta < 0. \end{cases}$$

Proposition 2.20 (A trace bound). *For every $X \in \mathfrak{sl}_2(\mathbb{R})$, $\operatorname{tr}\exp X = 2c(\delta) \geq -2$. Equality holds only when $\exp X = -I$. Consequently, an element of $\mathrm{SL}(2,\mathbb{R})$ is not an exponential if its trace is less than -2, or if its trace is -2 and it differs from $-I$.*

Proof. Since $\operatorname{tr} X = 0$, we have $\operatorname{tr}\exp X = 2c(\delta)$. For $\delta \geq 0$, $c(\delta) \geq 1$. For $\delta < 0$, $c(\delta)$ is a cosine, so it is at least -1.

Suppose $c(\delta) = -1$. Then $\delta < 0$, and $\sqrt{-\delta}$ is an odd multiple of π. Hence $\sin\sqrt{-\delta} = 0$ and $s(\delta) = 0$. Therefore $\exp X = c(\delta)I + s(\delta)X = -I$. ∎

Example 2.21 (A non-exponential). Let $a > 0$ with $a \neq 1$, and set

$$g_a = \begin{pmatrix} -a & 0 \\ 0 & -1/a \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}).$$

Its eigenvalues are negative and unequal, and

$$\operatorname{tr} g_a + 2 = -\left(a + \frac{1}{a} - 2\right) = -\frac{(a-1)^2}{a} < 0.$$

By the trace bound, Proposition 2.20, the element g_a is not $\exp X$ for any $X \in \mathfrak{sl}_2(\mathbb{R})$. For $a = 2$, the trace is $-5/2$.

The element g_a is a product of two exponentials. With $h = \operatorname{diag}(1,-1)$,

$$\exp(\pi J) = -I, \qquad \exp\big((\log a)h\big) = \begin{pmatrix} a & 0 \\ 0 & 1/a \end{pmatrix},$$

so $g_a = \exp(\pi J)\exp\big((\log a)h\big)$.

Exercise 2.4 proves that $\mathrm{SL}(2,\mathbb{R})$ is connected. So generation by exponentials, Proposition 2.11, applies, but one exponential does not suffice. Local invertibility, Proposition 2.10, is not contradicted. For a near 1, the element g_a is near $-I$, far from I. Lecture 6 shows that in a compact connected group every element is a single exponential.

Self-checks

1. Is $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ injective? Is $-I \in \mathrm{O}(1,1)$ of the form $\exp(uK)$?
2. Can a real matrix with negative determinant be $\exp X$ for a real matrix X? Does this contradict generation by exponentials, Proposition 2.11?
3. If $X, Y \in W$ and $[X,Y] = 0$, what is $H(X,Y)$? Is $\log(\exp X \exp Y) = X + Y$ for all commuting X, Y?

Answers. (1) No, since $\exp(2\pi J) = I = \exp 0$. Also no: $\operatorname{tr}(-I) = -2$, while $\operatorname{tr}\exp(uK) = 2\cosh u \geq 2$. (2) No: for real X, $\det\exp X = e^{\operatorname{tr} X} > 0$. There is no contradiction, because $\mathrm{GL}_n(\mathbb{R})$ is disconnected by the determinant sign. (3) Every later term is an iterated bracket, so $H(X,Y) = X + Y$. Still, take $G = \mathrm{SO}(2)$ and $X = Y = \pi J$. The product $\exp X \exp Y$ is I. For the local inverse $\log\colon V \to U$ of exp on $\mathrm{SO}(2)$, $\log I = 0 \neq 2\pi J$. Here $2\pi J \notin U$, since $\exp(2\pi J) = \exp 0$, $0 \in U$ and exp is injective on U.

Summary

Every continuous one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$ is $t \mapsto \exp(tX)$. For a general Lie group, flows of left-invariant fields define exp, a local diffeomorphism at 0. Homomorphisms intertwine exponentials, and connected groups are generated by exponentials. The identity component is an open and closed normal subgroup with the same Lie algebra. The adjoint representation satisfies $\mathrm{Ad}(\exp X) = e^{\operatorname{ad} X}$ and $\operatorname{ad} X(Y) = [X, Y]$, and elements with $[X, Y] = 0$ satisfy $\exp(X + Y) = \exp X \exp Y$. Near the identity, the BCH series computes multiplication from the bracket. Globally, exp need not be injective or surjective, and the Lie algebra $\mathfrak{g}$ does not determine the components of G.

Sources and hypotheses

This section records what the sources prove for the convergent BCH formula, Theorem 2.18.

Hall, Sections 5.3–5.5, proves the integral form of the formula for matrix Lie groups and X, Y near 0. Hall, Section 5.6, computes the series only through degree three.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves the convergent series for connected groups. This suffices, because the identity component of G is open, has Lie algebra $\mathfrak{g}$ and contains each curve $t \mapsto \exp(tX)$. Knapp's proof reduces to matrices with $\|X\| + \|Y\| < \log 2$. There the degree-n term of his logarithm series (B.24) has norm at most $c_n(\|X\| + \|Y\|)^n$, where $-\log(2 - e^s) = \sum_n c_n s^n$ and $c_n \geq 0$. Hence the series converges uniformly for $\|X\| + \|Y\| \leq r < \log 2$, and H is continuous. Knapp's Proposition B.43 converts the rational coefficients of (B.24) into rational bracket coefficients. Kirillov Jr., Section 3.7, Theorem 3.37, states the same series.

Reading

Read Hall, second edition, Chapter 2 on the matrix exponential, Chapter 3 on matrix Lie algebras and homomorphisms, and Sections 5.3–5.7 on BCH and group versus algebra homomorphisms. Kirillov Jr., Chapter 3, treats the general exponential map and the Campbell–Hausdorff formula. Warner, Chapter 3, Sections 3.46–3.47, treats the adjoint representation on a general Lie group. Lecture 3 uses the exponential to study subgroups and quotients.